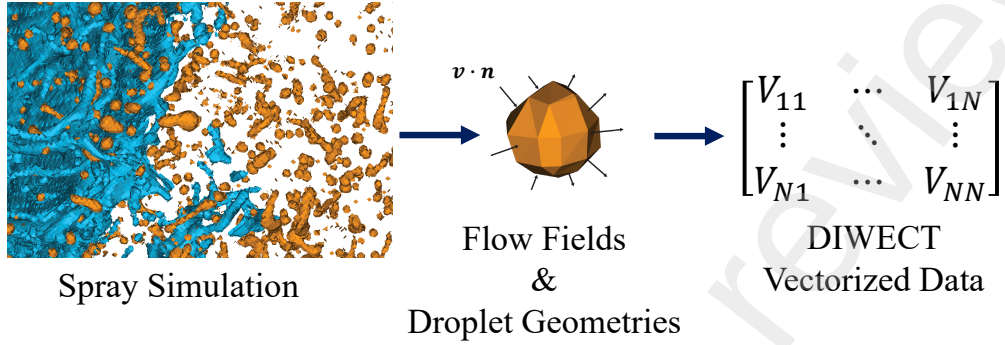




Graphical Abstract

Geometric and Topological Data Analysis of Atomization Simulations for Model Development

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Highlights

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- A novel method to extract droplet geometries and flow fields from atomization simulations is implemented.
- The DIWECT is described and applied to atomization data.
- It is shown that complex geometric data from simulations can be vectorized while retaining key physical characteristics.

Geometric and Topological Data Analysis of Atomization Simulations for Model Development

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Abstract

High-fidelity simulations of atomization present a promising approach to studying its complex physics. However, these simulations face significant challenges, often requiring hundreds of processors and generating tens to hundreds of terabytes of data, complicating their practical application. This work presents a method to extract localized geometries of droplets and their surrounding flow fields at the moment of breakup within a simulation, resulting in a comprehensive database of atomization statistics. This improves the usefulness of simulation results by storing key information on the atomization process. Parsing and comparing these geometries remains challenging, so we apply the directionally integrated weighted Euler characteristic transform (DIWECT) to produce vectorized representations of the complex data. These vectors are well-suited for advanced data processing, machine-learning applications, and the development of reduced-order models. We demonstrate the effectiveness of this approach through both shape matching and K-means clustering to confirm that the DIWECT captures meaningful physical information in an abstract, reduced-order format. This work represents a novel, efficient, and physically relevant method with which to extract, parse, and analyze large-scale atomization simulation data with an eye toward reduced-order model development.

Keywords: Physics extraction, data reduction, volume of fluid, geometric and topological data analysis

1. Introduction

Atomization—the process where a coherent liquid body breaks into smaller droplets—is a wide-ranging phenomenon with applications in numerous industries and environmental applications. From fuel injection and fire suppression to agricultural spraying and air-sea interactions, a detailed understanding of the underlying physics is essential for optimizing engineering designs and improving predictive models of natural phenomena. Despite its widespread importance, many of the underlying physical mechanisms that drive atomization remain unresolved. There remains significant research in the field attempting to answer the fundamental questions: when and how do liquids breakup?

Experimental studies on atomization have long faced limitations due to the inherent complexity of capturing high-speed, multiphase flows. Simplified jet flows can be captured well using traditional high-speed imaging techniques [1]. However, as the Weber and Reynolds numbers increase to values relevant for many practical applications, experimental analyses of the physics become intractable because of the formation of an opaque cloud of droplets that forms around the spray. Even with recent advances in imaging techniques like ballistic [2, 3, 4] and x-ray imaging [5, 6, 7], capturing temporally continuous, quantitative data remains problematic. As a result, experimental methods have struggled to provide sufficient insight into the small-scale instabilities and breakup mechanics that occur in atomizing systems.

In response to these challenges, high-fidelity simulations—commonly known as direct numerical simulations (DNS)—have emerged as a powerful alternative [8] capable of resolving the detailed physics of atomization. High-fidelity simulations continue to produce increasingly accurate representations of the complex multiphase process, providing researchers with a tool to predict quantitative, three-dimensional, and temporally granular descriptions of liquid breakup. However, these simulations are not without their own limitations. As computational power has increased, the size of datasets generated from atomization simulations has grown dramatically. High-resolution simulations, capable of producing hundreds of terabytes of data, present immense challenges in both storage and analysis. Extracting meaningful physics from such massive datasets remains an ongoing challenge, and current methodologies are ill-equipped to fully utilize this wealth of information.

To address these limitations, the Atomization Simulation Statistics Extraction Tool (ASSET) [9, 10] was developed to extract detailed event-based

data from large-scale simulations of atomizing liquids. Unlike traditional post-processing methods that are constrained by the sheer volume of data, ASSET works coincident with a simulation to identify and record breakup and coalescence events as they occur. This approach allows for the creation of comprehensive databases that contain statistics, geometries of liquid structures, and the corresponding flow fields at the moment of each breakup event.

Furthermore, ASSET tracks the evolution of each liquid structure, allowing for the reconstruction of breakup *pathways*. These are the sequential processes by which liquid structures fragment into progressively smaller droplets. Breakup pathways reveal how individual droplets emerge from larger structures, shedding light on the mechanisms that govern spray formation. By capturing these data directly from simulations, researchers gain an unprecedented level of access to temporally continuous, localized information essential for studying the full physics of atomization.

High-fidelity simulations are incredibly costly, often requiring hundreds of thousands of CPU hours on high-performance computers. Thus, predictive atomization models play a vital role, especially for practical applications where fully resolving the physics is not feasible. Reduced-order models such as Reitz-Diwakar (RD) [11], Taylor Analogy Breakup (TAB) [12], and Pilch-Erdman (PE) [13] models offer relatively fast and inexpensive ways to predict atomization behavior in specific systems. However, these models are heavily based on simplified assumptions and empirical data from idealized conditions, such as single droplets in controlled environments. As a result, they often fail to accurately represent the complex, multiscale behavior of atomization in real-world, turbulent systems. There is an increasing need for models that are informed by statistically relevant physics-based data extracted directly from simulations, enabling more accurate predictions of atomization dynamics across a broader range of conditions.

To meet this need, an essential step forward involves not only extracting more comprehensive data from simulations but also transforming these data into a format suitable for modern analysis techniques. Concretely, it is necessary to efficiently query and parse databases full of data from physical simulations. The present work proposes using a geometric and topological data analysis tool known as the directionally integrated weighted Euler characteristic transform (DIWECT) to convert complex geometric and flow field data from atomization simulations into vectorized representations. This approach structures the data in a format well suited for machine learning and other

data-driven techniques, enabling the development of reduced-order models that agree with the physics of atomization predicted by high-fidelity simulations. The resulting structured database enables efficient querying and analysis, supporting the development of reduced-order models that retain the underlying physics of atomization rather than relying on significant assumptions.

This paper focuses on demonstrating the process of generating these databases and applying DIWECT to enable future data-driven predictive models. Furthermore, the DIWECT provides a powerful foundation for novel analyses, enabling the identification of previously unrecognized patterns within these large-scale, complex systems. In doing so, we aim to bridge the gap between the vast amounts of data produced by high-fidelity atomization simulations and the limited models currently used, advancing toward a more accurate and physically informed understanding of atomization.

1.1. Background - Topological Data Analysis

Modern experimental and numerical methods in all range of scientific fields are producing increasingly complex and large-scale data, requiring efficient methods to analyze and visualize them. Topological data analysis (TDA) methods have emerged as a family of tools for summarizing and analyzing data in a mathematically robust and rigorous manner. For a comprehensive background on the various methodologies see *Topological Methods in Data Analysis and Visualization VI* [14]. In the context of fluid mechanics, where massive datasets are a common challenge, TDA offers the ability to capture key geometric components of complex fields.

There exist a few studies incorporating TDA into fluids [15, 16, 17, 18, 19, 20]. Scalar fields in fluid mechanics, such as pressure, temperature, and image pixel intensity can be analyzed using the persistent homology TDA method. These analyses have been used to identify large-scale symmetries, transition regions, and patterns in the flow [21, 22]. Similarly, studies involving vector fields, such as velocity, have used topological descriptors to classify coherent flow features and reveal interactions within turbulent flows [23, 24]. While these applications illustrate the versatility of TDA in fluid mechanics, they are focused on scalar field data and large-scale flow dynamics rather than capturing the small-scale, underlying physics of liquid breakup studied in this work.

In contrast, the geometric analysis of three-dimensional liquid structures in multiphase flows through TDA remains a relatively underexplored area.

Chen et al. [25] introduced a skeletonization technique to capture the fundamental topology of periodic liquid jets during atomization. This method allowed for the identification of branching points in the jet structure and offered new insights into the mechanisms driving liquid breakup. To the authors' best knowledge, this is the only topological analysis of liquid structure geometries in multiphase flows. Beyond fluid mechanics, TDA has been successfully employed in fields such as medical imaging [26, 27, 28], 3D object recognition [29], and structural biology [30] to quantify, match, and analyze complex shapes. These examples demonstrate the versatility of geometric TDA in capturing intricate structural information and motivate its application to fluid systems.

Building on these developments, this work focuses on analyzing three-dimensional geometries associated with liquid structures during atomization. Concretely, we use a topological tool referred to as the DIWECT for this purpose. Unlike previous methods that emphasize whole flow fields, the present work applies TDA to isolated liquid breakup events, providing concise, representative descriptions of complex geometries.

1.2. Background - Atomization Modeling

One of the primary objectives of this work is to provide physical data that can inform reduced-order atomization models. Achieving this requires an understanding of the information needed and the limitations of current modeling approaches. This section focuses on the role of Lagrangian droplet tracking models, as the data extracted from this study is particularly suited to their improvement.

The Lagrangian-Eulerian (LE) technique is a method used to model atomization with reduced computational cost. It represents the liquid phase as discrete, Lagrangian “blobs” or droplets, while the gas phase is treated as an Eulerian domain. Prominent implementations, such as the KIVA series [31, 32, 33, 34, 35], have been widely adopted in industry [36]. Following injection, the breakup of blobs is typically modeled using idealized instability mechanisms, such as Kelvin-Helmholtz and Rayleigh-Taylor instabilities [37, 11, 38]. Further breakup is often predicted using models like the Taylor Analogy Breakup (TAB) [12] or Pilch-Erdman (PE) [13]. These methods offer computational efficiency but rely heavily on simplifying assumptions, such as spherical droplet shapes and empirical correlations from isolated experiments.

These simplifications fail to capture the multiscale and turbulent nature of real-world atomization processes. Experiments and DNS have revealed that atomization involves complex deformation and breakup mechanisms, including bag and sheet formation, ligament destabilization, and droplet fragmentation [1, 39, 40]. These mechanisms lead to highly variable, complex droplet and flow field geometries that are far beyond the capabilities of traditional blob injection models.

Recently, there has been considerable development on the hybrid Volume-of-Fluid Lagrangian-Eulerian (VOFLE) method [41, 42, 43, 44, 45]. These models combine the Volume-of-Fluid (VOF) technique to resolve near field jet deformation and large-scale liquid structures before transitioning to an Lagrangian description for smaller droplets (Note: Level-set (LS) is also used for interface capturing in some cases [43]). Varying local criteria are used to determine when a droplet meets the requirements for Lagrangian treatment, with most opting for a size threshold for the drop diameter, and others additionally considering the sphericity of the droplet. Only two papers have considered further breakup to the author's best knowledge. These are Yu et al. [46], who consider the RT instability for breakup and Kuo & Trujillo [45], who use the maximum entropy formalism (MEF) model. VOFLE balances computational efficiency and accuracy with an improved resolution of the early stages of jet formation and breakup.

Despite these advancements, VOFLE models face challenges associated with empirical assumptions and computational trade-offs. For example, most methods require that droplets be spherical and/or below a size threshold before transitioning to a Lagrangian representation. This increases the necessary size and cost of the Eulerian domain. Additionally, secondary atomization is largely neglected, and when it is considered, secondary atomization models are significantly limited.

This work lays the foundation for a predictive VOFLE methodology, informed by high-resolution simulations. By extracting explicit breakup cascades, droplets can be transformed to Lagrangian structures at larger and less spherical sizes and their subsequent breakup can be more accurately predicted. The envisioned approach would identify liquid structures as they detach, compare them to a database of high-fidelity DNS results, and use stored breakup pathways to inform how a given structure will fragment. While this study does not implement the full VOFLE model, it establishes techniques for extracting droplet geometries, surrounding flow fields, and breakup genealogies, which are necessary steps toward improving atomization modeling.

2. Project Outline and Scope

This work seeks to establish a novel method to bridge the gap between high-resolution simulations and reduced-order modeling of atomization processes. The ultimate goal is to use the data and processing techniques outlined in this paper to inform reduced-order atomization models. However, the development of a model is beyond the scope of the present work. The current project is structured into two key objectives:

1. **Use ASSET to extract atomization geometries:** The ASSET methodology extracts the gas-liquid interface of liquid structures, along with their surrounding flow fields, from high-fidelity simulations during breakup and coalescence events. These structures and their interactions are stored in a graph database, allowing researchers to analyze the sequence of breakup events that lead to the formation of smaller droplets.
2. **Vectorize the resultant databases using the DIWECT:** The geometric and flow field data extracted using ASSET are transformed into vectorized representations with the DIWECT. This process prepares the data for machine learning applications, allowing for statistical analysis and efficient shape-matching. By capturing key geometric and topological features, the DIWECT helps identify patterns and relationships within the data.

This work provides an approach to analyze large atomization simulation databases, yielding information necessary to inform a physics-based reduced-order model. Explicitly, it is important to analyze the DIWECT as a method for vectorizing data in a *useful* way—where useful refers to the ability of these representations to serve as effective inputs to machine learning algorithms. An ideal vectorization must preserve the key geometric and topological characteristics of complex flow structures while reducing them to a comparable format suitable for statistical analysis and training. This ensures that meaningful similarities and differences between structures are captured in a way that algorithms can learn from and generalize across. This quality will provide both novel, in-depth, data-driven analyses into atomization as well as the potential to use these insights for predictive modeling.

3. Methods

3.1. Computational Platform

The advancements outlined below are built into the NGA computational platform [47, 48, 49, 50]. This platform is then used to perform the atomization simulation outlined below. NGA solves the two-phase, incompressible Navier-Stokes equations for mass and momentum conservation, expressed as

$$\frac{\partial \rho_\phi}{\partial t} + \nabla \cdot (\rho_\phi \mathbf{u}_\phi) = 0 \quad (1)$$

and

$$\frac{\partial \rho_\phi \mathbf{u}_\phi}{\partial t} + \nabla \cdot (\rho_\phi \mathbf{u}_\phi \otimes \mathbf{u}_\phi) = -\nabla p_\phi + \nabla \cdot (\mu_\phi [\nabla \mathbf{u}_\phi + \nabla \mathbf{u}_\phi^T]) + \rho_\phi \mathbf{g}, \quad (2)$$

where ρ_ϕ is the density, $\mathbf{u}_\phi$ the velocity vector, p_ϕ the pressure, μ_ϕ the dynamic viscosity, t the time, $\mathbf{g}$ the gravitational acceleration vector, and ϕ the phase indicator (g for gas and l for liquid). These equations are discretized on a staggered Cartesian grid, pressure stored at cell centers and velocity components at cell faces.

Numerical methods include:

- Time discretization: An iterative Crank-Nicolson scheme, with a semi-implicit correction [51]
- Gas-liquid interface tracking: A geometric volume-of-fluid (VOF) method with unsplit semi-Lagrangian fluxes [49]. The interface is reconstructed using piecewise linear interface calculation (PLIC) [52] with efficient least-squares VoF interface reconstruction algorithm (ELVIRA) for normal vector calculations [53] on subcells to ensure mass and momentum conservation [50].
- Advection: Away from the interface, high-order finite differences for mass, momentum, and scalar transport [47] are used. Near the interface the semi-Lagrangian VOF method is used [49].
- Pressure Solver: The Poisson equation is solved with the ghost fluid method to incorporate the surface tension force [54] and a black-box solver [55].

- Surface Tension: Curvature is computed using the adjustable curvature evaluation scale (ACES) method [56].

NGA is fully parallelized via MPI, enabling efficient scaling to tens of thousands of cores [57]. Its computational efficiency and robust numerical methods have made it an effective tool for studying atomization and related multiphase flow phenomena [58, 59, 49, 60].

3.2. ASSET Methodology

The first goal in this project is to extract liquid structure and flow field geometries from high-fidelity simulations. To meet this goal, it is necessary to identify key moments and extract only those structures that provide model-relevant data. The breakup process is important because it describes the formation of droplets. However, breakup represents only part of the dynamics in large-scale atomization. Coalescence is also common in dense spray clouds. Thus, tracking both breakup and coalescence is necessary to fully describe the system.

In addition, liquid structures often deform significantly between breakup or coalescence events. These deformations provide information on the mechanisms leading to breakup, such as an elongating ligament undergoing Plateau-Rayleigh instability or a droplet flattening before undergoing bag-type breakup.

The Atomization Simulation Statistics Extraction Tool (ASSET) is designed to capture this information by identifying and isolating breakup, coalescence, and intermediate (update) events. ASSET is an event-triggered, parallelized algorithm that operates concurrently with simulations. It runs each timestep and efficiently identifies breakup and coalescence events, gathering physical data associated with each occurrence. A comprehensive description of the algorithm is given in the original works [10, 9], however, a concise summary is included here. As we detail below, the key addition in this work is the ability to extract topologically sound geometries and to incorporate update events for more comprehensive tracking of liquid structure evolution.

The workflow for data extraction involves three key steps: event identification, geometry extraction, and velocity calculation on the extracted meshes. 1) Event identification works by monitoring changes in identification numbers, providing liquid structure history and breakup pathways. 2) Once an event is detected, the geometries of the affected liquid structures are extracted by isolating the computational cells involved in the event. To

generate a watertight surface representation, a marching cubes algorithm is applied, ensuring topological accuracy and avoiding artifacts that could interfere with subsequent analyses. 3) Finally, rather than storing the full velocity field, only the velocity component normal to the liquid surface is recorded, as this component is most relevant to droplet deformation. Details on each step are provided below.

3.2.1. Event Identification

Event identification in ASSET (described in detail by Christensen and Owkes [9]) begins by tagging liquid structures with two unique integers: $\mathcal{S}$ and $\mathcal{L}$. $\mathcal{S}$ is assigned at every timestep using a band-growth algorithm [42], distinguishing independent liquid structures within the domain. $\mathcal{L}$ is persistent across timesteps and is transported with the liquid, only changing when breakup or coalescence occurs. $\mathcal{L}$ provides a history of the liquid, allowing us to track the breakup pathway of any given liquid structure.

The event identification algorithm distinguishes the three types of events as follows:

- **Breakup Events:** Occur when a single liquid structure (same $\mathcal{L}$) splits into multiple, structures (different $\mathcal{S}$) in non-adjacent grid cells.
- **Coalescence Events:** Identified when two or more structures with distinct $\mathcal{L}$ values flux into a single grid cell.
- **Update Events:** Triggered periodically for each structure based on a user-defined time interval relative to the last event the structure was involved in.

The original method described the extraction of statistics when an event was identified. This utility is still applied. These data are sampled both immediately preceding and following an event, providing a record of the conditions leading to and resulting from each event. Data are stored in a Neo4j graph database (see Figure 1), enabling relationships between parent and child droplets to be established and liquid structures to be tracked back to the initial coherent liquid at the start of the simulation.

3.2.2. Geometry Extraction

When an event is identified, the region around the liquid structure(s) in the event is isolated. In fact, this is already done during the assignment of $\mathcal{S}$

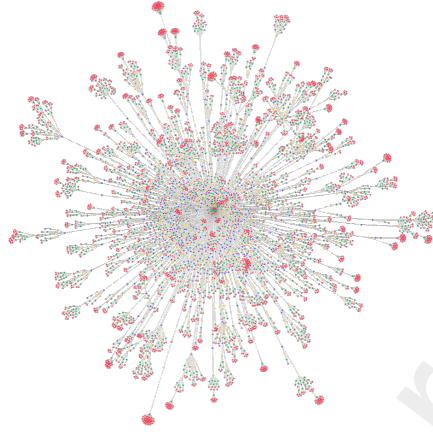


Figure 1: Neo4j graph database representing data extracted from a numerical simulation of a diesel-type jet from [9]. Dots represent droplets with lines connecting related droplets (i.e parent and child droplets). Colors represent the number of breakups taken to get to a given droplet.

through a band-growth algorithm [42], providing access to the computational cells of interest. With the cells isolated, the liquid-gas interface geometry is captured. Ensuring no erroneous topological features are captured in this step is necessary because the DIWECT methodology is highly sensitive to them. A naive implementation would be to output the PLIC representation of the interface, which is already calculated. However, as seen in Figure 2a, the edges and vertices of the PLIC interface are not aligned to produce a watertight surface, introducing non-physical holes. To ensure topological soundness, we implemented a formulation of the commonly used marching cubes algorithm [61] that is topologically robust [62]. This method generates a set of triangles representing the interface. The vertices and connectivity information of these triangles are stored as an unstructured mesh in the SILO format [63]. In this format, each triangle is represented as a zone, and scalar values can be associated with these zones. The marching cubes algorithm provides a topologically sound representation of the droplet, however, it does not ensure a fully accurate volume, whereas the PLIC interface is volume conserving, which is necessary for the geometric volume-of-fluid numerical methods. Producing a volume-conserving, water-tight representation and preventing the calculation of two separate interface reconstructions is an area of future work.

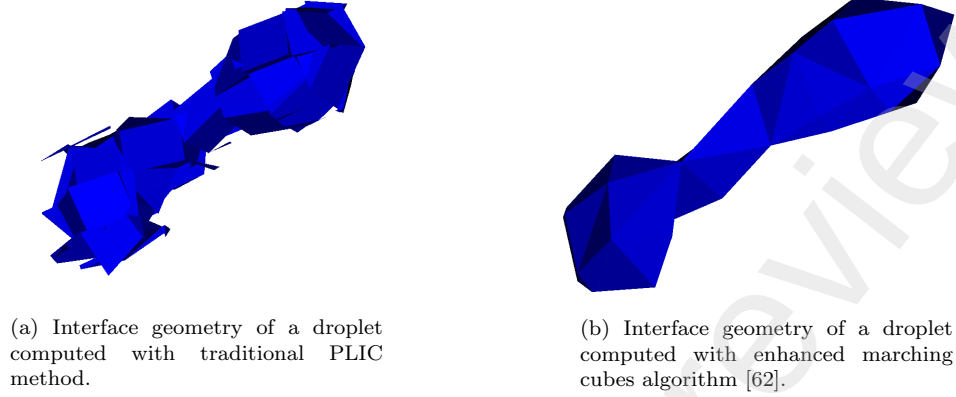


Figure 2: Comparison of the volume conserving PLIC interface and the topologically sound, watertight marching cubes interface.

3.2.3. Velocity Calculation

Saving the full velocity field around each droplet would present a significant storage requirement. To reduce the total data being saved, the flow field is simplified to include only the velocity component normal to the interface. This normal component, represented mathematically as $\mathbf{u} \cdot \mathbf{n}$, where $\mathbf{u}$ is the velocity vector and $\mathbf{n}$ is the unit normal vector of the interface plane, is most prominently responsible for droplet deformation.

Since $\mathbf{u} \cdot \mathbf{n}$ is a scalar quantity, it can be efficiently stored in the SILO file for each triangle zone. With the orientation of the interface plane and the velocity magnitude, the interaction between the velocity and the shape of the fluid is efficiently stored together. The simplified data can then be visualized and analyzed using tools such as VisIt [63], enabling detailed studies of droplet dynamics.

3.3. The Directionally Integrated Weighted Euler Characteristic Transform (DIWECT)

With droplet geometries and their associated normal velocities now extracted and stored efficiently in a compact, structured format, the next objective is to develop a means of comparing these complex shapes and dynamics in a consistent, scalable manner. Specifically, we require a vectorized representation that captures the essential geometric and flow-related features of each droplet, enabling fast and meaningful comparison across a large database of breakup events. The Directionally Integrated Weighted Euler Characteristic Transform (DIWECT) is introduced here as a method to provide exactly

this vectorization, translating high-dimensional surface and velocity data into forms suitable to statistical analysis and machine learning.

This section details the methodology of the DIWECT. This method is an extension of the weighted Euler characteristic transform (WECT). For a more in-depth background on the WECT, see Jiang et al. [64] and Cisewski-Kehe et al. [65].

The foundation of this method are simplicial complexes. Explicitly, for a set of vertices $\mathcal{V} \subseteq \mathbb{R}^n$, a *simplex* is the convex hull of an affinely independent subset of $\mathcal{V}$. Given any simplex σ , we define its dimension $\dim \sigma$ as the number of vertices in σ minus one (e.g. a zero-simplex is a vertex, a one-simplex is an edge, a two-simplex is a triangle, etc.). A face τ of a simplex σ is any subset of its vertices ($\tau \leq \sigma$), while σ is a coface of τ ($\sigma \geq \tau$). A *geometric simplicial complex* is a finite collection of simplices K satisfying:

1. If $\sigma \in K$, then all faces $\tau \leq \sigma$ are also in K
2. If any two simplices $\sigma, \sigma' \in K$, have a nonempty intersection, then $\sigma \cap \sigma'$ is a face of both σ and σ'

A *weighted* simplicial complex (K, ω) is defined by a simplicial complex K and a weight function $\omega : K \rightarrow \mathbb{R}$, that assigns a weight to each simplex in K . The unstructured meshes extracted in §3.2.2 are simplicial complexes with the normal velocities calculated in §3.2.3 as weights on each simplex. Weights for the edges and vertices are calculated by averaging the weights of their respective co-faces.

The Euler characteristic of a simplicial complex is given as

$$\chi(K) = \sum_{\sigma \in K} (-1)^{\dim \sigma}. \quad (3)$$

For example, for three-dimensional polyhedra ($\dim 3$), the Euler Characteristic is defined $\chi = V - E + F$ where V , E , and F are defined vertices, edges, and faces, respectively. For example, the Euler characteristic of any Platonic polyhedra (e.g. tetrahedron, cube, octahedron, etc.) is two. The Euler characteristic is a topological invariant, meaning it remains unchanged under continuous deformations such as bending or stretching, as long as there is no tearing or gluing.

The Euler characteristic is easily extended to the *weighted Euler characteristic* (χ_ω) for weighted simplicial complexes, defined as

$$\chi_\omega(K, \omega) = \sum_{\sigma \in K} \omega(\sigma) \cdot (-1)^{\dim \sigma}. \quad (4)$$

The weighted Euler characteristic alone is a limited representation of any complex, because it reduces all information to a single scalar value. However, by computing it across a sequence of subcomplexes, we can retain more information from the data. This is done using a *filtration*, which is a nested sequence of weighted simplicial complexes parameterized by $\mathbb{R}$. Given a filtration $\{K_t\}_{t \in \mathbb{R}}$, it satisfies

$$K_s \subseteq K_t \text{ for all } s \leq t. \quad (5)$$

This method specifically uses a *height filtration*, which orders simplices based on their vertex positions in a given direction $d \in \mathbb{D}^{n-1}$. The height filtration of a complex $K_{d,t}$ is defined as

$$K_{d,t} := \{\sigma \in K \mid h_d(\sigma) \leq t\}, \quad (6)$$

where $h_d(\sigma)$ is the maximum height of any vertex in σ along direction d , which is computed using the standard inner product of $\mathbb{R}^n$. In the WECT, we take the height filtration in multiple directions, thus providing a structured and mathematically rigorous way to represent complex geometries as vector-valued quantities. See Figure 3 for a demonstration of this process.

By computing the Euler characteristic over a height filtration $\{K_d\}_{t \in \mathbb{R}}$ of a weighted simplicial complex (K, ω) for a direction $d \in \mathbb{D}^{n-1}$, we obtain the *weighted Euler characteristic function* ($\text{WECF} : \mathbb{R} \rightarrow \mathbb{R}$). This function maps a direction d and a height t to the weighted Euler characteristic of the corresponding subcomplex $K_{d,t}$. Formally

$$\text{WECF}_{(K,\omega),d}(t) := \chi_\omega(K_{d,t}, \omega). \quad (7)$$

Finally, computing the weighted Euler characteristic function for all directions $d \in \mathbb{D}^{n-1}$, we define the *weighted Euler characteristic transform* ($\text{WECT} : \mathbb{D}^{n-1} \times \mathbb{R} \rightarrow \mathbb{R}$) as the function

$$\text{WECT}_{(K,\omega)}(d, t) := \text{WECF}_{(K,\omega),d}(t). \quad (8)$$

Theoretically, the WECT fully encodes the shape of the simplicial complex K , when computed over all directions d and heights t . In practice, the WECT is computed for a finite sampling of directions and heights. Nevertheless, with sufficiently dense sampling, the WECT retains enough information to uniquely characterize a given simplicial complex [64, 65].

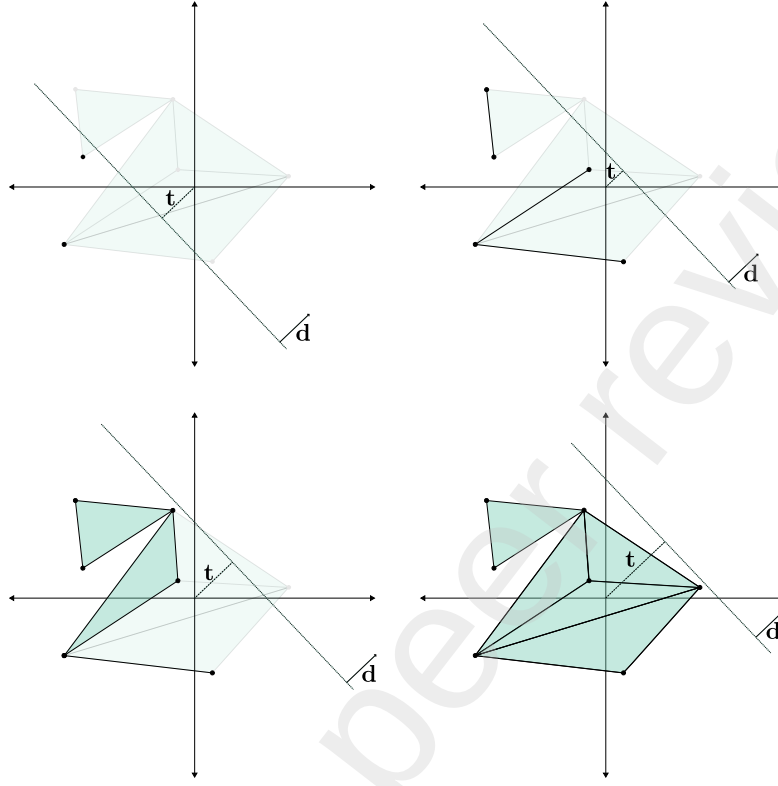


Figure 3: Height filtration in direction d . Simplices become bold when they are below the plane and contribute to the Euler characteristic for that height t .

The WECT is sensitive to rotations and reflections of the simplicial complex. To address this limitation, we introduce the *directionally integrated* WECT, or DIWECT which integrates the WECT over all rotations and reflections of the simplicial complex.

For a uniformly sampled set of directions $\mathcal{D} \in \mathbb{D}^{n-1}$, the DIWECT can be estimated as a directional average of the WECT and computed as

$$\text{DIWECT}_{(K,\omega)}(t) = \frac{1}{|\mathcal{D}|} \sum_{d \in \mathcal{D}} \text{WECT}_{(K,\omega)}(d, t). \quad (9)$$

In practice, we are not able to compute the WECT over all directions and heights, so we use an estimated form of the DIWECT, and the theoretical guarantees are necessarily limited by this discrete sampling. For this study, we selected 150 directions and 75 heights to balance two competing objectives: preserving fidelity of the droplet representation and minimizing

storage requirements. As shown in Equation 9, the directional sampling is averaged over each height, yielding a 75-dimensional vector per droplet. This dimensionality was chosen to ensure that the resulting representation remains compact and do not increase the storage requirement for small droplets. For example, an octahedron, which is among the most simplified and lowest-resolution shapes extracted in our dataset, requires approximately 50 integers to represent its vertices, faces, and associated velocity components. Most droplets in the dataset are significantly more complex than an octahedron, so a 75-element vector does not substantially increase the storage burden for the majority of cases, while still providing a decently high-resolution description of shape variation across directions. Moreover, since the DIWECT of a single orientation is sufficient to represent any rotated or reflected transformation of a droplet, we avoid the need to store multiple transformed variants, effectively reducing the total storage cost by a factor of $\mathcal{D}$, the number of sampled directions.

Algorithm 1 Methodology Summary

```
1: Run Simulation and Extract Data with ASSET
2: while simulation is running do
3:   Assign initial  $\mathcal{S}$  and  $\mathcal{L}$  identifiers in the domain
4:   for each timestep do
5:     Reassign  $\mathcal{S}$  for each liquid structure
6:     Transport  $\mathcal{L}$  with the liquid
7:
8:     Event Identification:
9:     if Multiple  $\mathcal{S}$  associated with one  $\mathcal{L}$  then
10:       Breakup event identified
11:     end if
12:     if Multiple  $\mathcal{L}$  in one cell then
13:       Coalescence event identified
14:     end if
15:     for each structure do
16:       if (Current time % update frequency = 0) then
17:         Update event identified
18:       end if
19:     end for
20:
21:     Geometry Extraction:
22:     for each identified event do
23:       Extract the gas–liquid interface via marching cubes
24:       for each interface cell do
25:         Velocity Calculation:
26:         Compute  $\mathbf{u} \cdot \mathbf{n}$ , on each triangle of the mesh
27:       end for
28:       Save as unstructured mesh with normal velocities
29:     end for
30:   end for
31: end while
32:
33: Post-Processing: Estimate the DIWECT
34: Aggregate all extracted data
35: for each unstructured mesh  $\chi$  do
36:   Estimate the DIWECT of  $\chi$ 
37: end for
38: Output: A vectorized representation of liquid structure geometries from
    atomization
```

4. Results

The results presented here are derived from $\approx 120,000$ droplets extracted from a high-fidelity simulation run. The entire simulation database, made up of a full rendering of each timestep requires $\approx 10\text{TB}$ of storage space. The total storage requirement of the unstructured SILO meshes was 133GB. The full WECT was calculated with 150 directions and 75 heights and only required 1.5 GB of storage. The averaging step was conducted over each direction and the DIWECT representation of the same dataset only required 10 MB—a reduction of over 13,000-fold from the SILO meshes and 1,000,000-fold from the original simulation dataset.

The following subsections provide proof of concept of the DIWECT as a method to uniquely and efficiently characterize liquid shapes and surrounding flow fields. First, we match droplets to their most similar counterparts using only their DIWECT values. Then, we show that the droplets can be clustered into groups which display physical relevance based on their DIWECTs.

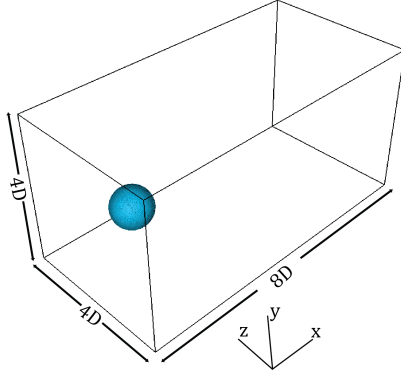
4.1. Simulation Setup

To test the methodology, we ran a large-scale numerical simulation of an atomizing water droplet (Fig. 4a). The methods outlined in this work are designed to be scalable and capable of processing large numbers of droplets, as would be expected in most practical atomization applications. The setup in this work consists of a high-speed gas inflow at the x^- wall, which is added to an initialized homogeneous isotropic turbulence (HIT) domain and inflow, as shown in Figure 4b. This simulation ran for one month on 128 processors on the Tempest HPC at Montana State University. To reduce the size of the resultant dataset, only droplets from the second half of the simulation were analyzed.

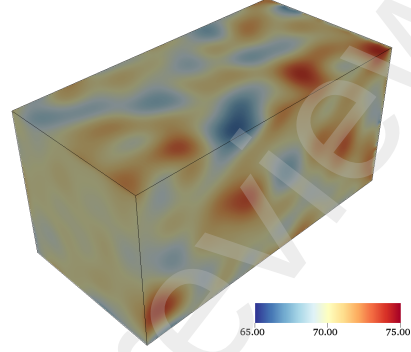
This test case represents a common atomization scenario with broad applications, including aerial firefighting, agricultural spraying, industrial spray cooling, and environmental aerosols. The chosen resolution balances computational efficiency with sufficient detail to capture smaller-scale droplets. The key non-dimensional parameters for this simulation are listed in Table 1.

4.2. Droplet Matching

A key objective of this work is to establish a basis for efficiently comparing droplets and their local flow fields across large simulation databases. In a future scenario, a user might extract a particular droplet of interest



(a) Simulation domain with initial droplet in place (D is droplet diameter).



(b) HIT-derived turbulence initialized in the domain.

Figure 4: Simulation domain setup.

Table 1: Non-dimensional parameters for the simulation. D is initial droplet diameter, dx is grid spacing, ρ_g is gas density, U_g is the bulk gas velocity, μ_l is liquid viscosity, σ is the surface tension coefficient

Parameters	Description	Value
Cells across diameter	D/dx	150
Bulk Reynolds	$\rho_g U_g D / \mu_l$	4.74×10^4
Bulk Weber	$\rho_g U_g^2 D / \sigma$	825

from an ongoing simulation and automatically search a previously computed database for the most similar droplet. Then, predictions can be made from the previously simulated droplet to inform how the droplet of interest might behave.

For demonstration, we selected a set of four droplets from a separate simulation run. These droplets were pulled from another water drop atomization case, but with a larger initial diameter. Four common shapes from the database were chosen. Each droplet's DIWECT signature was then compared against the database developed from the simulation in §4.1 to find the most similar entries according to Euclidean distance in DIWECT-space.

These results can be seen in Figures 5-8, which show the reference droplet on the left, and the closest matched droplets on the right. AR is the aspect ratio, defined as the shortest principal axis divided by the longest, and V , which is the volume of the droplet.

The matching results show very nice qualitative agreement for the shapes. There exists some variation between the shape parameters, but it is important to note that there are many interacting factors that lead to the final DIWECT calculation, notably, the DIWECT captures information about the normal velocity components. Thus, comparing a simple shape parameter and the size of droplets is a limited assessment of the DIWECT's capability of to compare similar structures.

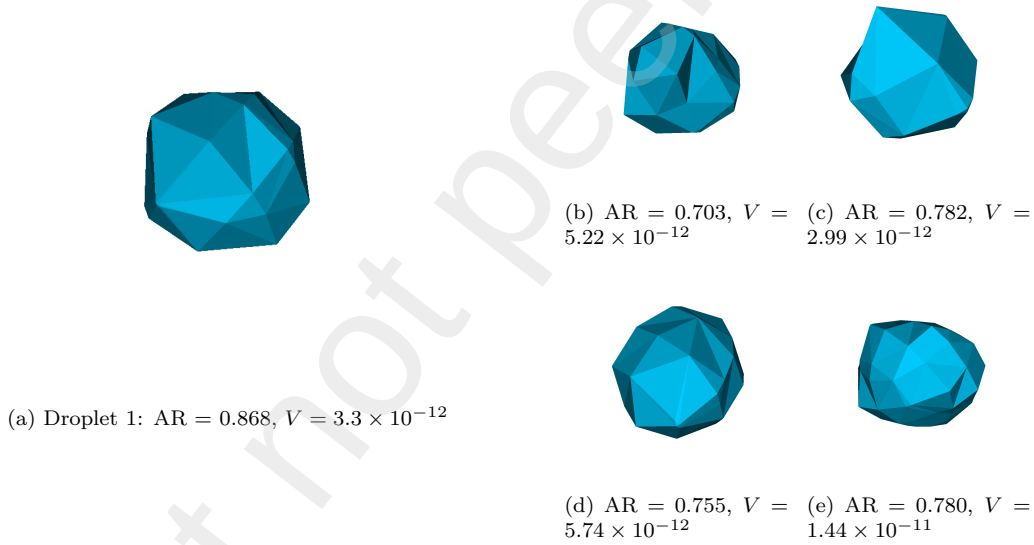


Figure 5: Droplet 1 and the four closest matches based on DIWECT

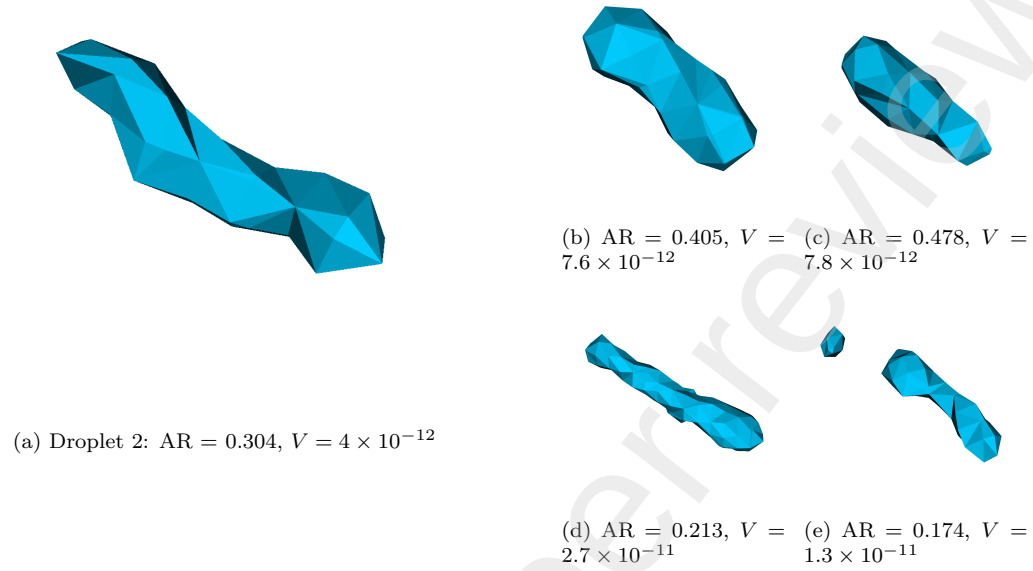


Figure 6: Droplet 2 and the four closest matches based on DIWECT

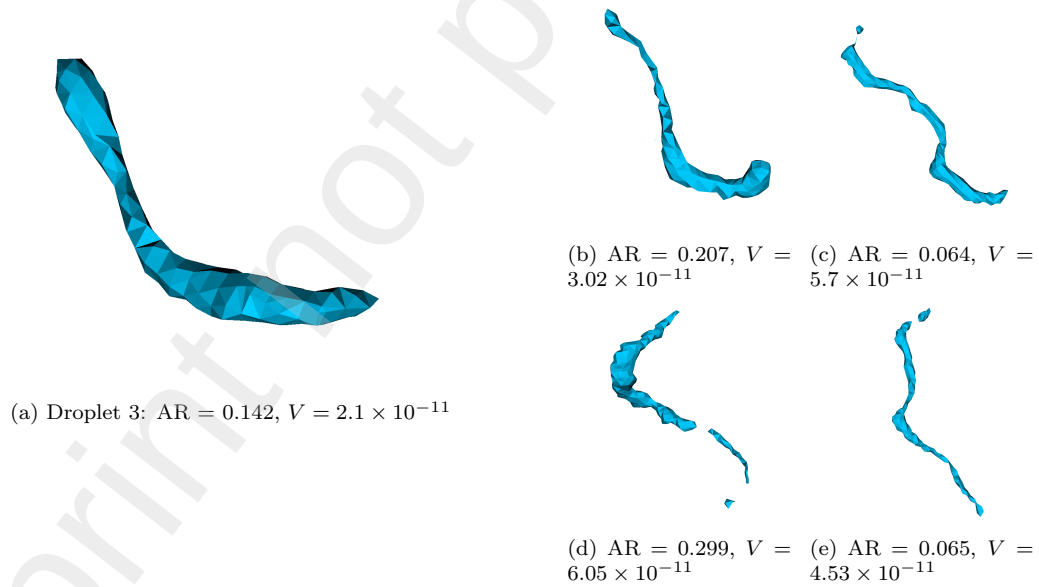


Figure 7: Droplet 3 and the four closest matches based on DIWECT

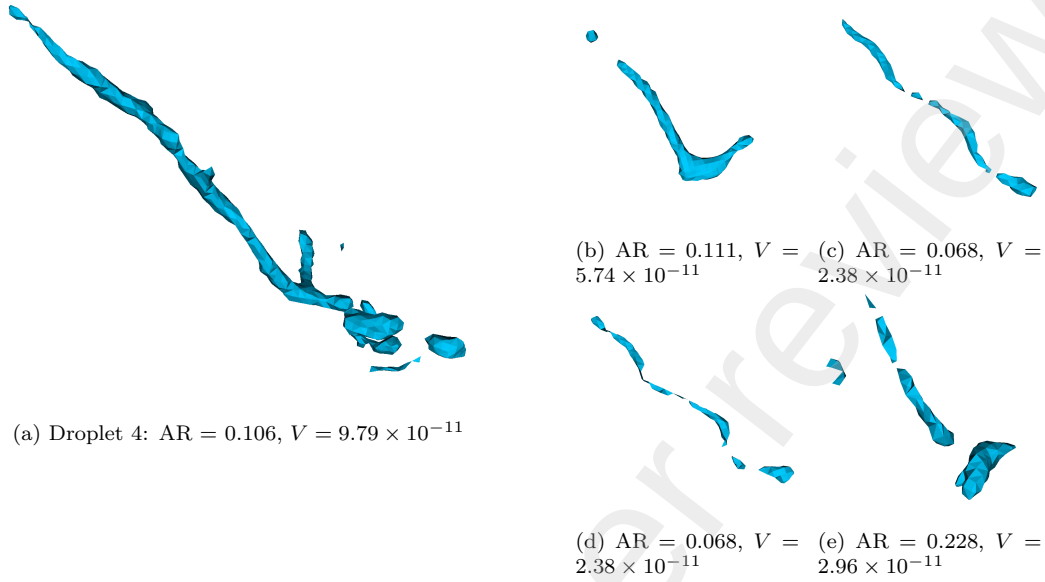


Figure 8: Droplet 4 and the four closest matches based on DIWECT

4.3. Clustering Results

To further demonstrate the utility of the DIWECT, we apply clustering to the resulting vectors using Euclidean distance as a measure of similarity. By comparing the resulting clusters with known physical parameters, we assess how effectively the DIWECT preserves relevant physical information. Given the large size of the dataset ($\approx 120,000$ samples), computational efficiency is a key consideration. To ensure efficiency of our analysis at scale, we consider methods better than $O(n^2)$ where n is the number of samples. Another consideration is the dimensionality of the data. The discretization produces a database of 75-dimensional vectors, so the dimensionality must be considered to avoid sparse sampling in high-dimensional space, otherwise known as the curse of dimensionality. Projecting the data into a lower dimension improves the capability of non-parametric methods and enhances computational efficiency. Principal Component Analysis (PCA) is therefore applied to the dataset before clustering. We found that 20 components was sufficient to explain the variability of our data (See Figure 9). To determine the appropriate number of PCA components, we used the elbow method, which identifies the point beyond which additional components contribute diminishing returns in explained variance. Following the application of PCA, we cluster the data

using the K-means algorithm [66]. We choose K-means because of its scalability, with a complexity of $O(nki)$, where n is the number of droplets, k is the number of clusters, and i is the dimensionality after PCA. This methodology provides both computational efficiency and a systematic way to assess, in a quantitative manner, whether the DIWECT representation retains physical information from atomization systems.

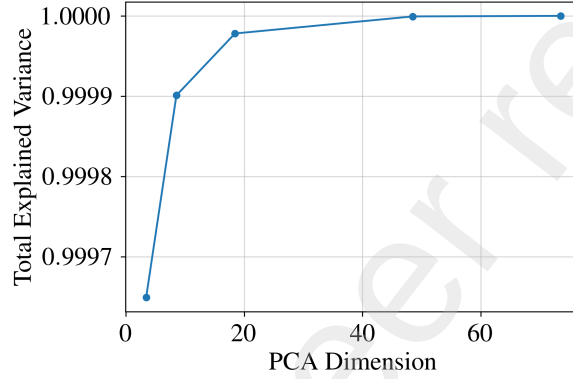


Figure 9: Total displayed variance of data with varying PCA dimensions.

We begin with relatively few clusters ($K = 4$) to illustrate how the droplets naturally separate into groups with similar morphologies. Figure 10 provides a random sampling of droplets from each cluster. Some droplets appear similar because they are either observed at consecutive update events (when only minor shape changes occur) or they represent large structures that experience a small breakup event (e.g. droplet shedding off of a ligament), which is not visible at the scale of the image.

From inspection of Figure 10, cluster 0 appears to contain small, elongate droplets, whereas cluster 1 has small but more spherical droplets. Clusters 2 and 3 generally contain larger, more complex shapes. A selection of statistics describing these clusters is shown in Figure 11, with mean values and standard deviations indicated by error bars. The quantities plotted—volume, aspect ratio, slip velocities, and cluster population—capture important physical characteristics of the breakup process. Aspect ratio is defined as the ratio of the smallest principal axis length to the largest principal axis length (L_{min}/L_{max}); lower values indicate more elongate shapes, while higher values indicate more spherical droplets. Slip velocities refer to the difference between the droplet velocity and the surrounding gas velocity in the x-, y-, and

z-directions, which influences the aerodynamic forces acting on the droplet. The slip velocity is a slightly different (though related) quantity than the normal components of the velocity used as weights in the DIWECT. The slip velocity is calculated as a weighted sum of the gas velocity upstream of the droplet, which should represent the portion of the flow that impinges on the droplet.

Figure 11 shows that clusters 0 and 1 are composed of large numbers of small droplets; cluster 0 tends to have more elongate droplets (lower aspect ratio), while cluster 1 contains droplets that are more spherical. In contrast, clusters 2 and 3 contain fewer but larger structures, with cluster 3 showing the greatest mean volume. Slip velocities do not vary drastically among clusters, suggesting that size and shape features contribute more toward the similarity or difference of droplets' respective DIWECT values.

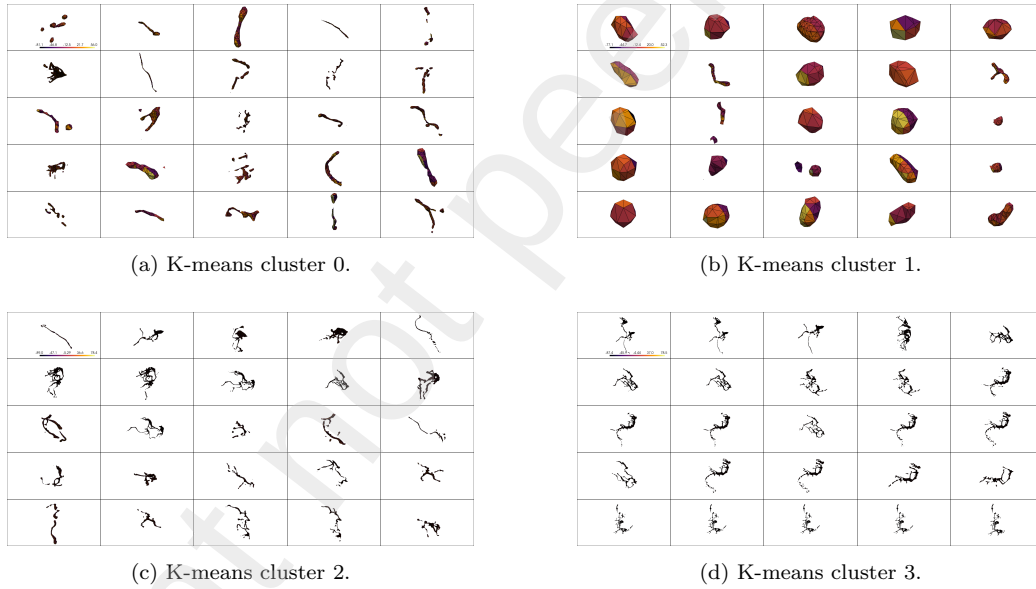


Figure 10: Visualization of the droplets in K-means clusters for $K = 4$.

Next, we increase the number of clusters to analyze. This will provide further analysis of how the DIWECT differentiates physical features. In the $K = 4$ clusters, it was clear that the size and shape of droplets were ultimately the key features in differentiating them (i.e. see Figs. 11a and 11b). While these are key factors in atomization, velocity contributes significantly to breakup mechanisms. The effects of velocity should become more appar-

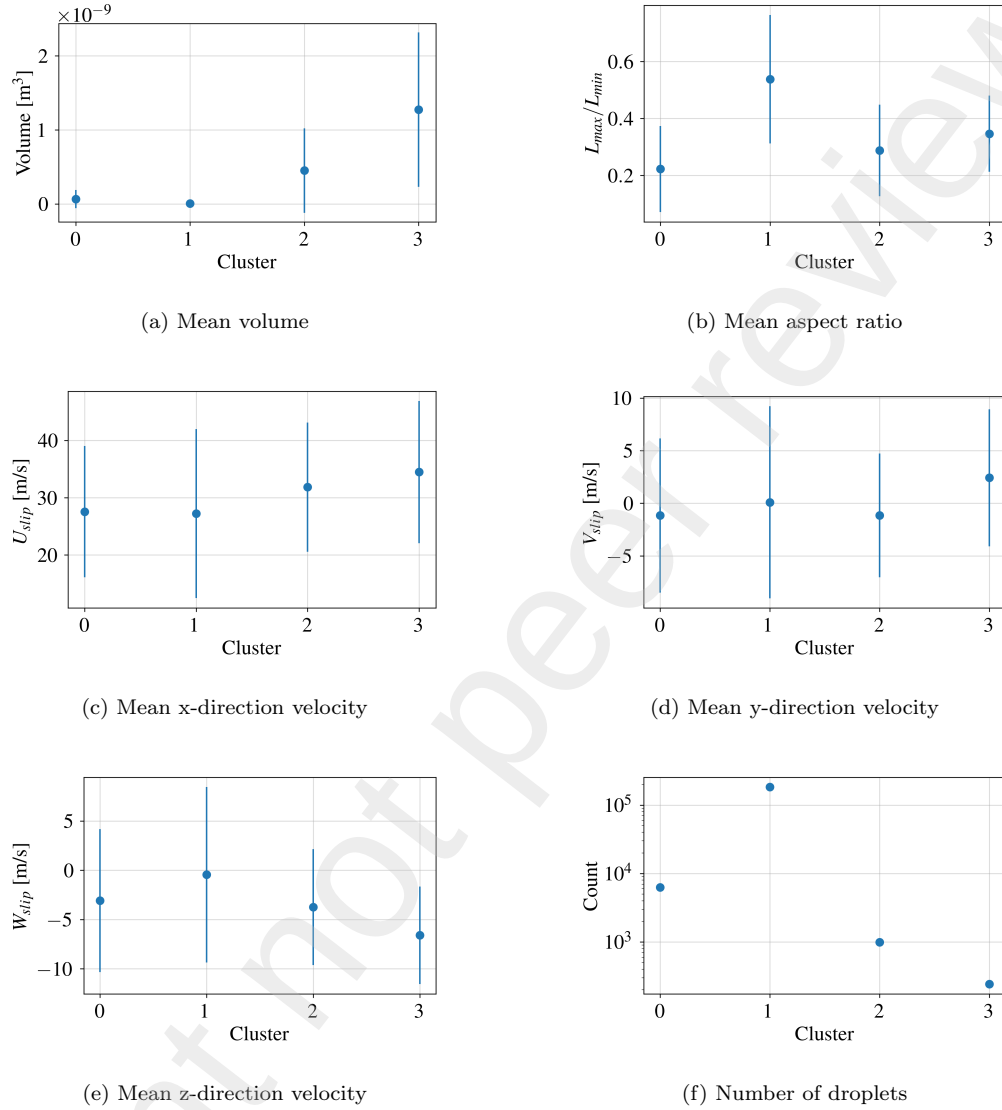


Figure 11: Statistics for clusters created by k-means clustering.

ent, as there are enough clusters to capture droplets with similar shapes, but differing local velocity fields. This will provide further confidence that the WECT can successfully differentiate structures by not only geometry, but also by their surrounding flow conditions.

The results for $K = 16$ are shown in Fig. 12. The large cluster 1 in the

original $K = 4$ collection is now spread out among clusters 1, 9, and 14, which display small-volume droplets with relatively high aspect ratios (see Fig. 13 for visual confirmation). Among these, cluster 14 exhibits a lower mean aspect ratio compared to clusters 1 and 9 (≈ 0.5 vs. ≈ 0.6). While clusters 1 and 9 appear geometrically similar, they were clustered separately, likely suggesting an influence from velocity. Notably, cluster 9 displays a stronger x-direction slip velocity than cluster 1 ($\approx 35\text{m/s}$ vs. $\approx 25\text{m/s}$), suggesting that droplets in this grouping were subjected to differing flow conditions, potentially experiencing higher shear. This differentiation implies that variations in velocity fields contribute to different breakup behaviors, which are now captured by the greater number of clusters.

5. Conclusions

This work has taken two significant steps toward a more comprehensive analysis of atomizing flows. First, we have shown that it is possible to extract liquid shapes and associated flow fields from high-fidelity simulations in a way that directly targets breakup and coalescence events. Previous approaches generally rely on full-field outputs [57, 67, 68, 23], but the method presented here isolates each relevant structure and captures localized information at the moment of breakup. This capability represents a shift from broad snapshots of the entire simulation domain to data that provides insights into model-relevant dynamics.

Second, we have demonstrated a means of transforming these complex, high-dimensional datasets into vectorized representations using the DIWECT. By approximating a mapping of geometry and local velocity fields to concise vectorized descriptors, we can readily apply machine learning or other data-driven tools to an otherwise intractable volume of simulation data. Our clustering examples illustrate that droplets sharing similar shape, size, and flow conditions appear in common groups, reaffirming that the DIWECT captures physically meaningful features.

While these contributions move atomization research closer to fully data-driven reduced-order modeling, there are limitations that must be addressed in future work. ASSET's requirement that distinct liquid structures be separated by a full grid cell can sometimes cause the output geometries to misrepresent connected components, either by grouping individual structures together or artificially separating continuous ones. This issue is evident in Figs. 10 and 13 and presents a challenge for the DIWECT, which is sensitive

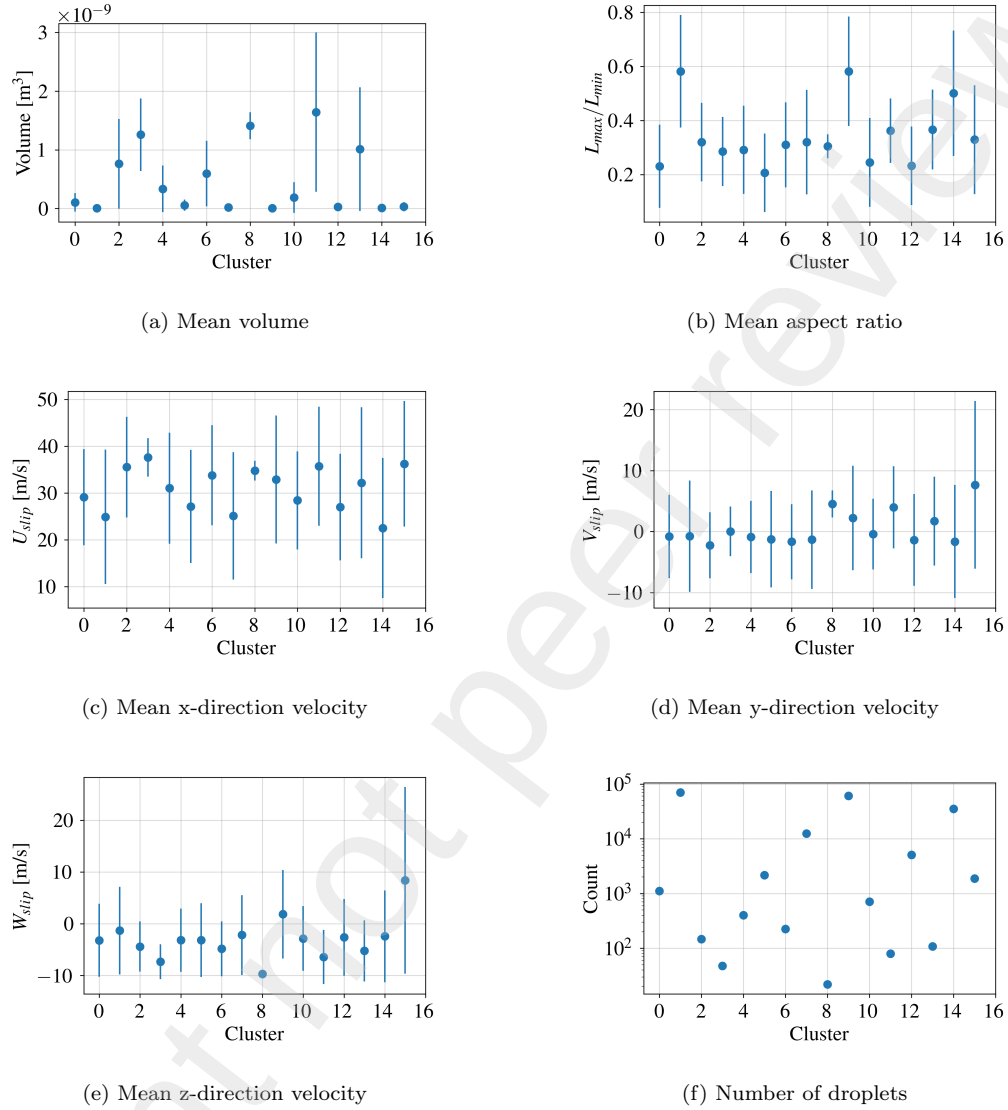


Figure 12: Statistics for clusters created by k-means clustering.

to discontinuities in interface representation. This artifact can be reduced with increasing simulation resolution, which is possible with continually improving numerical methods and computational efficiency. However, more robust treatment of liquid structures at the moment of breakup would improve the methodology. Implementations of numerical methods such as

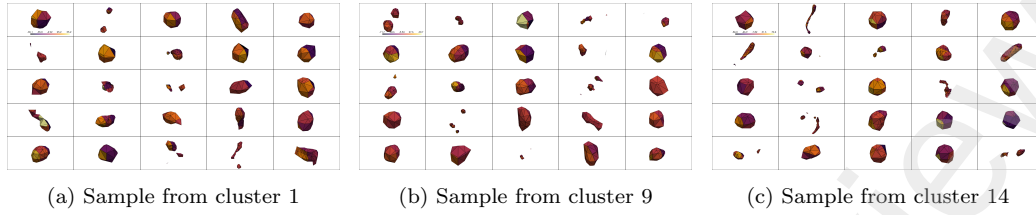


Figure 13: Renderings of samples of droplets from clusters 1,9, and 14 in $K = 16$ clustering output.

reconstruction with 2 planes (R2P) [69] in output simulations would greatly improve these artifacts.

Despite the limitations, these methods show promise that physical data can be recovered from the DIWECT. As noted in §3.3, given an infinite number of sampling directions and heights, the DIWECT is a one-to-one mapping of any shape. Practically, the method must be discretized, so we chose 150 directions and 75 heights for the present application, then an estimate of the DIWECT was computed. This resulted in a data reduction from 133 GB to 10 MB. Applying the methodology presented in this work to a predictive tool will necessitate tuning these sampling parameters, however this remains outside the scope of the present work.

The key results in this work show that the DIWECT can be used for finding the closest match which can determine the closest match to a given droplet in a database of droplets. This will be useful as a breakup model where the database can be used to predict the dynamics of a droplet. Additionally, clustering was performed by first computing PCA on the DIWECT database to further reduce the data size and to combat the curse of dimensionality. The clustering results first demonstrate the utility of the DIWECT as a way to represent droplets by classifying them into logical bins, and second to highlight the ability of the DIWECT to aid in analyzing simulation results where droplets can be clustered to understand the number of droplets in various types.

Ultimately, this work bridges the gap between massive simulation datasets and meaningful data for model development. The methods outlined here represent significant improvements in the field of atomization simulations, by allowing greater utilization of massive, costly simulation runs. Future work will focus on model development to create a novel category of physics-based models without significant empirical assumptions.

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